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Thermal-Aware Qubit Mapping Optimization

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DATES

Since the advent of quantum computing, scientists have studied how to efficiently perform high-level calculations using the power of quantum superposition and entanglement. One area that has been intensely researched is developing an architecture for quantum computers that would allow for reliable execution of algorithms, without being limited by the physical limitations of the current NISQ devices. While there are several different methods by which the compilation of quantum circuits can be accomplished, most of the current architectures, such as SABRE's, do not account for the thermal characteristics of the physical qubits, resulting in significant levels of decoherence. This study examines the performance of a thermal-aware qubit mapping algorithm that minimizes SWAP overhead while maximizing the thermal stability of all physical qubits in the circuit. The proposed thermal-aware qubit mapping strategy utilizes a cost function to achieve a balance between minimizing the SWAP overhead and a maximizing the thermal stability of the physical qubits. A key component of this technique is the use of a generated graph to initialize the mapping algorithm, with both Qiskit and NetworkX working together to accomplish this task. We also compared the circuit fidelity of the proposed thermal-aware qubit mapping against both random and SABRE based mapping and showed that the thermal-aware mapping achieved a circuit fidelity of 0.974, and a higher degree of thermal stability than the SABRE circuit mapping (0.968) despite requiring only one additional SWAP operation, demonstrating that thermal-reliability is an important and actionable element within quantum circuit optimization.

1. Introduction

With the beginning of the 21st century, quantum computers have emerged as one of the most promising and efficient areas of modern day technology. A classical computer processes information using a binary system (0's and 1's) as machine-level code, meanwhile; quantum computers operate using quantum bits (qubits) which can exist in multiple states or levels due to the characteristics of superposition and entanglement. These properties of qubits allow the quantum computers of today to solve problems that classical machines could not, including optimization, (cryptography) the secure transfer of information and simulation.

Unfortunately, the construction and operation of a quantum computer has proven very difficult. Presently, researchers are in the NISQ (Noisy Intermediate-Scale Quantum) era which refers to the current type of quantum computer that is constructed with limited amounts of qubits (the current state of quantum chips) and that are highly susceptible to errors (e.g., due to environmental noise) associated with the operation of qubit-based quantum computing. Thermal decoherence is one of the most significant sources of environmental noise affecting a quantum computer's operations. Thermal decoherence occurs when the quantum state of a qubit is destroyed by ambient heat. Operating multiple qubits on the same chip creates varying levels of thermal decoherence, making it nearly impossible to create a truly reliable qubit computer since all of the qubits would need to operate at the same temperature.

In real-world applications of quantum circuits, qubit mappings occur as part of the process of assigning logical qubits

to physical qubits in the hardware of a quantum computer. When a logical qubit requires information stored in another type of logical qubit that is far away, additional SWAP gates need to be added to facilitate the movement of quantum information, as the qubits in a quantum computer do not connect with one another directly. The algorithm that is most commonly used to accomplish this task is called "SABRE" (SWAP-based Bidirectional heuristic search), which is typically able to produce an efficient sequence of operations and minimize the number of SWAP operations. However, SABRE and the vast majority of other transposition algorithms are unable to assess whether qubits are currently running hot or cold, or whether they are likely to have high levels of thermal noise or low reliability, and so they are "thermally blind" to data that could impact the results of the mapping decisions.

The investigation into the shortcomings of this methodology has inspired us to develop a thermally aware qubit mapping algorithm that first identifies the coolest and most reliable (stable) physical qubit on the hardware to perform quantum operations before proceeding to the routing stage by way of SDQs. Our algorithm will utilize a combination of two different cost functions to enable decision makers to reduce both the SWAP overhead and the thermal noise associated with completing a mapping decision, thereby ensuring that critical operations will not be mapped to unreliable qubits. The algorithm has been developed for implementation in Python using Qiskit and NetworkX and will be demonstrated using an example of 5-qubit system using simulation.

2. Problem Formulation

Before presenting our solution, it is important to clearly understand the three core problems that exist with standard qubit mapping approaches on real quantum hardware.

2.1 Sparse Connectivity and SWAP Overhead

Real quantum hardware does not allow every qubit to directly interact with every other qubit. Instead, qubits are connected in a specific pattern called coupling map or hardware topology. For example, in our 5-qubit test device, qubits are connected along certain edges - meaning a gate between two non-adjacent qubits cannot be executed directly.

To handle this, compiler inserts SWAP gates to move quantum information across the chip until the required qubits are adjacent. However, each SWAP gate is itself decomposed into 3 CX (CNOT) gates, which are expensive and error prone operations. Therefore, more SWAPs that are inserted, the noisier and less reliable the circuit becomes. Minimizing SWAP count is thus a primary goal of any qubit mapping strategy.

2.2 Thermal Heterogeneity Across the Chip

Not all qubits are of the same quality, which is a problem that isn't talked about as much. On a real quantum chip, the temperature of each qubit is slightly different. This causes a lot of differences in:

- Some qubits make mistakes 2–5 times more often than others.
- T1 time is the amount of time a qubit can stay in its state before losing energy. On our test device, it ranges from 80μs to 150μs.
- T2 time is the time it takes for a qubit to stay in phase coherence (between 50μs and 110μs).
- Stability score: a way to rate how reliable a qubit is overall.

In the 5-qubit model, Qubit q2 (temperature - 10K, stability - 0.97) is much more stable and reliable than Qubit q3 (temperature - 20K, stability - 0.78). Important tasks might end up using the worst qubits. If these procedures are ignored, which could result in mistakes that are fatal.

2.3 Thermal Blindness in Standard Transpilers

SABRE is the current industry-standard mapping algorithm that does a great job of lowering the number of SWAPs. But it doesn't use any information about noise or heat when it decides which qubit to assign first. It treats all physical qubits as equally reliable, which is not true for NISQ hardware.

This means that even a circuit that is routed perfectly can have low fidelity just because it was mapped onto unstable, hot qubits. This is a real gap in the tools we have for compiling code.

2.4 Formal Problem Statement

There is a quantum circuit C that has a set of logical qubits and gate operations.

A hardware topology graph G that shows how physical qubits are connected.

A thermal noise profile with T_1 , T_2 , and stability scores for each qubit, as well as the temperature and error rate for each qubit.

Goals: Find a mapping M (logical $\rightarrow$ physical) for the first qubit and a routing strategy that lowers the total cost:

Minimise $= \alpha \times \text{SWAP_cost} + \beta \times \text{Thermal_noise_cost}$

$\alpha = 0.6$, which is the weight given to how well the network connects and routes.

$\beta = 0.4$: The amount of weight given to thermal noise and qubit stability

The mapping should give the coolest and most stable physical qubits first to logical qubits that are used often.

This formulation includes both the goal of standard transpilers to be more efficient at routing and the lack of thermal awareness in them.

3. Proposed Approach: Thermal-Aware Mapping

This part talks about the solution we came up with for the qubit mapping problem. The main idea is simple: first, assign the most important logical qubits to the coolest and most stable physical qubits. Then, on top of that thermally optimised initial layout, use standard routing.

3.1 Main idea

Standard mapping algorithms, such as Sabre, only care about reducing the number of SWAP gates needed to route a

circuit. This is important, but it doesn't take into account where on the chip those operations will actually run. If the qubits in a circuit are mapped onto noisy, thermally unstable physical qubits, the circuit can still give bad results, even if it has very few SWAPs can produce poor results if it's qubits are mapped onto noisy thermally unstable physical qubits.

We solved this by treating qubit mapping as a two - objective optimization problem:

1. Minimize routing overhead (SWAP count)
2. Maximize qubit quality (thermal stability)

Both objectives have brought together and combined into a single unified cost function that helps the mapping process.

3.2 The Unified Cost Function

The core of the algorithm is based on the following cost function:

$$\text{Cost} = \alpha \times \text{SWAP_cost} + \beta \times \text{Thermal_noise_cost}$$

SWAP_cost : The estimated number of SWAP gates required for the mapping.

Thermal_noise_cost : Total noise contribution of the physical qubits selected for the mapping.

$\alpha = 0.6$ gives weight to connectivity efficiency.

$\beta = 0.4$ this gives significant weight to thermal quality.

α and β values were selected based on the experimentation to show that map routing is being prioritised but thermal noise cannot be ignored as they are important. These weights can change on the basis of device and circuit type.

3.3 Score Calculation for Stabilization

For every physical Qubit, we calculate a stabilization score that combines key noises characteristics:

(I) **Temperature (K)** - Low values are better

(II) **Gate error rate** - Low values are better

(III) **T1 time (μ s)** - higher values signifies high coherence

(III) **T2 time (μ s)** - higher values signifies high coherence

On the following table Qubits are ranked on decreasing order based on their stability rate. Here, 5 qubit test device is shown ;

Qubit	Temp (K)	Error Rate	T1 (μ s)	T2 (μ s)	Stability
q2	10	0.001	150	110	0.97 ✓ Best
q0	12	0.002	120	80	0.92
q4	15	0.0025	110	75	0.88

q1	18	0.003	95	60	0.85
q3	20	0.004	80	50	0.78 x Worst

According to the above table Q2 is most preferred Qubit for use and Q3 must be avoided for most of the tasks.

3.4 Thermal Noise Modelling

Calibration information for each qubit, such as gate error rates, stability scores, operating temperatures, time to decay (T1), and time to dephased (T2), is stored in a labelled data structure. Each qubit's thermal noise score can be derived through a weighted sum (equal) of normalized temperature and normalized error rate according to this equation:

$$\text{Thermal Noise Score} = 0.50 \times \text{Normalized_Temperature} + 0.50 \times \text{Normalized_Error_Rate}$$

A higher thermal noise score means that the qubit will have less reliability and be noisier. The thermal noise score is a direct input into the mapping cost function detailed in Section 3. In addition to determining the thermal noise scores, a complete noise model simulating thermal relaxation errors is created. All single gate and two gate error sources are modelled separately to accurately reflect instances of actual hardware behaviour since longer duration two qubit gates tend to have proportionally higher decoherence effects.

3.5 Algorithm and Analysis

The following algorithm presents the procedure of proposed thermal aware qubit mapping approach. The following algorithm takes as input the quantum circuit, hardware topology graph and thermal noise of every qubit profile and gives most efficient (Optimized) logical to physical mapping as result / output.

ALGORITHM:

Input: $C \leftarrow$ Quantum circuit with n logical qubits

$G \leftarrow$ Hardware topology graph with physical qubits Q

$T \leftarrow$ Thermal noise profile {temperature, error_rate, T1, T2, stability}

$\alpha \leftarrow$ Weight for SWAP cost (0.6)

$\beta \leftarrow$ Weight for thermal cost (0.4)

Output: $M \leftarrow$ Optimal mapping {logical qubit $\rightarrow$ physical qubit}

1: procedure THERMALAWAREMAP(C, G, T, α, β)

```

2:  PHASE 1: Thermal Profiling

3:  for each qubit  $q \in Q$  do

4:     $\text{norm\_temp}(q) \leftarrow (\text{temp}(q) - \text{min\_temp}) / (\text{max\_temp} - \text{min\_temp})$ 

5:     $\text{norm\_error}(q) \leftarrow (\text{err}(q) - \text{min\_err}) / (\text{max\_err} - \text{min\_err})$ 

6:     $\text{noise\_score}(q) \leftarrow 0.5 \times \text{norm\_temp}(q) + 0.5 \times \text{norm\_error}(q)$ 

7:  end for

8:   $\text{THERMAL\_ORDER} \leftarrow \text{sort } Q \text{ by } \text{stability\_score}(q) \text{ in descending order}$ 

9:  PHASE 2: Circuit Analysis

10:  $\text{logical\_edges} \leftarrow \text{extract all two-qubit gate pairs from } C$ 

11: PHASE 3: Candidate Generation

12:  $\text{candidates} \leftarrow \text{empty list}$ 

13: if  $n \leq 5$  then

14:   for each permutation  $P$  of  $Q$  taken  $n$  at a time do

15:      $M\_temp \leftarrow \{i \rightarrow P[i] \text{ for } i = 0 \text{ to } n-1\}$ 

16:      $\text{swap\_cost} \leftarrow \text{ESTIMATESWAPS}(M\_temp, \text{logical\_edges}, G)$ 

17:      $\text{thermal\_cost} \leftarrow (1/n) \times \sum \text{noise\_score}(P[i])$ 

18:      $\text{candidates.append}(\{M\_temp, \text{swap\_cost}, \text{thermal\_cost}\})$ 

19:   end for

20: else

21:    $\text{candidates.append}(\text{GREEDYMAP}(C, G, T))$ 

22: end if

23: ——— PHASE 4: Best Mapping Selection ———

24: sort candidates by (swap_cost ASC, thermal_cost ASC)

25:  $\text{best} \leftarrow \text{candidates}[0]$ 

```

```

26: PHASE 5: Routing

27: initial_layout ← best.mapping

28: transpiled_C ← SABRE_TRANSPILE(C, G, initial_layout)

29: PHASE 6: Evaluation

30: ideal_counts ← SIMULATE(transpiled_C, noise_model = None)

31: noisy_counts ← SIMULATE(transpiled_C, noise_model = T)

32: fidelity ← COMPUTEFIDELITY(ideal_counts, noisy_counts)

33: return {best.mapping, count_swaps(transpiled_C),
          avg_noise_score(best.mapping), fidelity}

34: end procedure

35: procedure ESTIMATESWAPS(M, logical_edges, G)

36: swap_count ← 0

37: for each edge (l0, l1) ∈ logical_edges do

38:   p0 ← M[l0], p1 ← M[l1]

39:   if (p0, p1) ∉ G.edges then

40:     swap_count ← swap_count + (dist(p0, p1) - 1)

41:   end if

42: end for

43: return swap_count

44: end procedure

45: procedure GREEDYMAP(C, G, T)

46: M ← empty mapping

47: available ← all physical qubits in Q

48: for each logical qubit i in C do

```



```

49:  best_physical ← argmin_{q ∈ available}
      (α × swap_cost(M ∪ {i→q}) + β × noise_score(q))

50:  M[i] ← best_physical

51:  available ← available \ {best_physical}

52: end for

53: return M

54: end procedure

```

END OF ALGORITHM

Complexity Analysis:

Phase	Complexity
Thermal Profiling	$O(n)$
Candidate Generation ($n \leq 5$)	$O(n!)$
Candidate Generation ($n > 5$)	$O(n^2)$ Greedy based
SABRE Route	$O(n^2 \times d)$ d = circuit depth
Overall for small circuits	$O(n!)$
Overall for large circuits	$O(n^2 \times d)$

3.5 Qubit Mapping Strategies

This study describes and compares three different approaches to mapping.

The Proposed Thermal-Aware Mapping Algorithm: All possible assignments of logical qubits to physical qubits for the 5-qubit circuit will be evaluated. All candidates for this mapping, as determined by the mapping procedure, will then be ranked by their respective SWAP overhead, and then by their score based on the Unified Cost Function described above which considers the tradeoffs between SWAP overhead and thermal noise. If two assignments require an equal number of SWAPs, then the thermal noise score will be used as tie-breaker. Qiskit's algorithm handles actual routing when the assignment which has lowest combined cost. Here proposed algorithm and SABRE works differently. Here proposed algorithm decides where operations has to take place inside the chip and SABRE decides how to route

them. Because of this we are free from implementing a full transpiler from scratch and still we have advantage of thermal placement without using any additional tools.

SABRE Mapping (Industry Baseline): The SABRE transpiler in Qiskit is used with default settings. SABRE reduces use of SWAP count by performing near optimal routing configuration but it does take important factors like thermal noise or Qubit stability in considerations. SABRE is represented as standard method in current industries for quantum circuit compilation.

Random Mapping (Adversarial Baseline): Here routing distance is maximized on a linear topology graph by assigning logical qubits in reversed physical qubit mapping order. This done to produce a lower bound performance by giving worst case number of SWAP gates. This done to give a realization how inefficient a thermally blind, unoptimized mapping can be when compared to both SABRE and proposed approach.

3.6 Fidelity Evaluation

In this work, Hellinger Circuit fidelity evaluation is used a primary performance metric.

Hellinger Fidelity:-

$$F(X, Y) = \left(\sum_i \sqrt{p_i q_i} \right)^2$$

In computation, fidelity is computed by running the transpiled circuit twice. First on an ideal simulator and then on thermal noise through QiskitAER simulator. The resulting output probability distributions of both simulators are compared. If a fidelity value of 1.0 is obtained, it means thermal simulation has perfect match with ideal simulator output. lower values ($1.0 > x > 0.0$, x being fidelity value) greater noise induced error. The evaluation metric a reliable measure of mapping quality which goes beyond the counting SWAP gates as it captures both effects, SWAP overhead and thermal noise.

3.7 Tools and Libraries

The table given below gives tools and libraries used in the implementation of this system.

Library	Purpose
Qiskit	Construction of Quantum circuit, transpilation and routing
QiskitAER	Noisy circuit simulation using thermal relaxation error models.
NetworkX	Hardware topology graph construction and shortest path computation
NumPy	Numerical operations and noise score normalization

Pandas	Loading and managing per qubit thermal calibration data
Matplotlib/ Seaborn	Generating comparison plots for SWAP count, noise score, and fidelity.

Table 2: Libraries

4. Experimental Evaluation

In this section we are showing experimental setup, results, and qubit mapping strategies analysis in this work. These experiments were conducted as a sample 5 qubit quantum device using QiskitAer simulators and test circuit which are described in section 4. Here using three metrics, performance is measured:

- SWAP gate count
- Average thermal noise score
- Circuit Fidelity.

4.1 Experimental Setup

The experiment uses a 5 - qubit stimulated quantum device with a partially connected linear topology, as shown in *figure 1*. In topology temperature of operating qubits (nodes) are represented as color, such as blue node indicate cooler qubits which are more stable, whereas red nodes indicate hotter and more unstable qubits.

Qubit 2 is coolest and it is most stable. Qubit 3 is operating in highest temperature and carries greater noise instability. Proposed mapping is motivated through this thermal heterogeneity across the chip.

The test circuit created for this work consist of 5 CX (CNOT) gates placed between non - adjacent qubits. Through this it is guaranteed that all three mapping strategies must some SWAP gate to overcome hardware connectivity constraints. To ensure reproductibility, using Qiskit every mapping strategy is transpiled with a fixed random strategy. Fidelity is computed using Hellinger fidelity evaluation, by comparing the output of ideal noiseless simulation and output of noisy simulation from the per - qubit thermal calibration data.

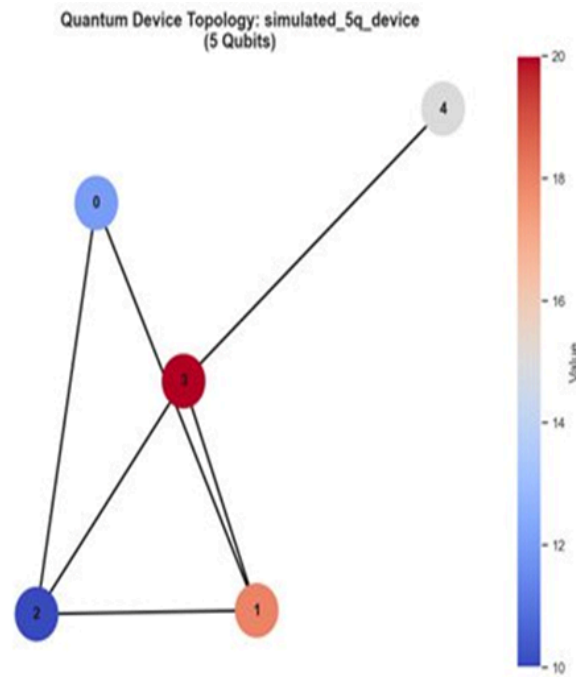


figure 1: Topology Graph

4.2 Results

In Table 3 given below shows numerical results for all three mapping strategies.

Method	SWAP_count	Noise Score	Fidelity
Random	7	0.510	0.951
SABRE	1	0.510	0.968
Thermal - Aware	2	0.510	0.974 - best

Table 3: Comparison of mapping strategies across all three performance metrics

4.3 SWAP Gate count Analysis

Here figure 2 shows the SWAP gate count for each mapping strategy. As we can see, Random mapping has maximum SWAP count of 7. This confirms that thermally blind and unoptimized placement leads inefficient and severe routing

overhead. Each SWAP gate results in 3, hence random strategy results in 21 additional gate operations into the circuit. Hence, a significant error is generated on NISQ hardware.

The way we put things together in a circuit matters. If we do not do it correctly it can create a lot of work for the circuit. Every time we use a SWAP gate it actually becomes three CX gates. This means that if we just put things together randomly we end up with 21 operations. This is a problem because it can cause a lot of mistakes on the hardware we are using.

SABRE is a method that does a job of reducing the number of SWAP gates. It only needs one SWAP gate. The aware approach needs two SWAP gates. This is more than SABRE. The reason for this is that the aware approach puts the qubits first. This can make the layout not perfect for connections. As a result we need one step. The results show that this is worth it.

4.4 Thermal Noise Score Analysis

Figure 3 shows the noise score for each way of mapping. It is interesting that all three methods have the noise score of 0.510. We need to be careful when we look at this result. The noise score is an average of all the qubits that are used during mapping. Since we use all five qubits in every strategy the average will always be the same. This does not mean that thermal placement does not matter. It just means that the average noise score is not an enough measure to see the difference between strategies. The fidelity metric is a way to evaluate the quality of mapping. It looks at the order of operations. Which qubits they are run on.

4.5 Fidelity Analysis

Figure 4 shows the comparison of fidelity. This is the important result of our experiment. The aware approach has the fidelity of 0.974. It does better than SABRE and the random method. The random method has a fidelity of 0.951. This is because it has a lot of SWAP operations. The extra SWAP operations cause a lot of mistakes. They make the circuit output worse. SABRE does better than the method. It has a fidelity of 0.968. This is because it reduces the number of SWAP gates. However SABRE does not consider which physical qubits are stable during placement. This means that some operations may still run on qubits. As a result SABRE's fidelity is limited.

The aware approach has the fidelity of 0.974. Even though it uses one SWAP gate than SABRE it produces better fidelity. This is because it assigns the used logical qubits to the coolest and most stable physical qubits. The reduction in mistakes from qubit selection makes up for the extra SWAP gate. This result shows that thermal placement quality affects circuit fidelity.

4.6 Key Insight

The important thing we learned from these results is that reducing SWAP count is not enough to maximize circuit fidelity. SABRE has SWAP gates than the thermal-aware approach but it produces lower fidelity. This shows that the quality of the qubits we select during placement is just as important, as the number of SWAP gates. The aware approach does a job of capturing this by treating placement and routing as a single problem.

5. Discussion

5.1 Results Analysis

The results presented in Section 5 corroborate this work's main thesis: qubit placement informed by knowledge of device temperature statistics allows routing compilers to produce circuits with measurably better fidelity on NISQ hardware, even at the cost of higher SWAP count. Because of thermal-aware placement, our strategy achieves 0.974 fidelity whereas SABRE achieves 0.968, a difference of 0.006. Though these numbers do not seem large out of context, it should be noted that we are working with a fidelity gap of just over half of a percent. Current NISQ devices are noisy enough that the difference of a few thousandths of fidelity can mean the difference between a correct output and an incorrect output when run on near-term hardware. These results clearly show that this difference does exist.

An interesting point that arises in Section 5.4 is that all three routing strategies achieved equal noise scores of 0.510. Though this makes sense mathematically because the noise score is simply an average of all assigned qubits' individual error rates, a shortcoming of this becomes clear: taking an equal-weighted average of all assigned qubits does not specify which physical qubits are executing the largest number of gates. This makes sense when thinking about the goal of noise scoring, which is to represent as accurately as possible the effective noise that the input circuit will experience when run on a given hardware configuration. Because the noise score does not take into account how many gates are executed by each qubit, it fails to properly weight each qubit's individual noise scores by the total number of gates it will execute. Future work should address this limitation when designing an evaluation framework.

5.2 Justification of SWAP Count Increase

The reader might be wondering how the thermal-aware strategy produces better results than SABRE if it uses two SWAPs instead of one, as SABRE did. The reason comes down to how errors accumulate in real quantum hardware. Each SWAP gate will undoubtedly introduce error due to the three CX gates that it uses to swap qubits, but if those CX gates are executed on physically cool qubits with lower error rates and higher coherence times, then the cumulative error introduced will still be quite small. Compare this to running a single gate on a physically hot qubit with unfavorable properties; in many cases, that one operation will introduce more decoherence than three CX gates on much cooler, well behaved qubits. The thermal-aware strategy does not avoid this trade-off, it just makes it explicit. By using two SWAP gates instead of one, we are able to guarantee that all logical gates will be run on the lowest-error, coolest physical qubits. The results speak for themselves and show that this is indeed the trade-off we should be making.

5.3 Related Work

Quantum circuit routing transpilers like SABRE were originally designed with a single-minded purpose: to reduce the circuit SWAP count as much as possible. More SWAPs means longer routing paths and a longer runtime on hardware. This was not necessarily a bad design decision when noise parameters across quantum hardware were relatively homogeneous and poorly understood. As time has gone on and backend teams have started providing calibration data on a per-qubit basis as standard practice, temperature variation across qubit arrays has become increasingly extreme. At the time of writing, the heavy/light physical qubit distinction used in this work would be considered par for the course on any of the leading devices tracked by metrics developed by the Quantum Architecture Roadmap Program [6]. Results developed in this work

support claims by many others in the literature that noise-aware compilation consistently outperforms noise-unaware compilation with respect to output fidelity. This work takes a step in that direction by showing that something as simple as a thermal-aware initial layout paired with stock SABRE routing can produce better results than stock SABRE.

5.4 Future Work

While promising, there are several shortcomings to this work that must be acknowledged. Firstly, this experiment was run on a simulated 5-qubit device. Though the noise model used by the simulator was built from sampled thermal calibration parameters, a simulator cannot perfectly model all of the sources of error that arise in a real quantum device. Factors like crosstalk between adjacent qubits and time-variant fluctuations in local qubit error are difficult to predict. Secondly, this work only implements the experiment on a single 5-qubit routing pressure circuit. It would be more convincing to users if the approach was tested on a large variety of circuits of different sizes, depths, and gate configurations. Thirdly, this work's thermal-aware mapper uses an exhaustive permutation search to generate valid mappings. Such an approach does not scale to circuits larger than five qubits. If this methodology is to be applied to larger circuits and devices, some heuristic search strategy would need to be employed. Fourth, as discussed in Section 6.1, the noise score metric used to evaluate each strategy could be improved.

6. Discussion

6.1 Integration with Existing Quantum Hardware

An easy way to deploy this project to real quantum hardware is if the target device makes per-qubit calibration data available to users. IBM Quantum devices provide users with a daily calibration report for each of their devices which includes not only T1 and T2 times for each qubit, but readout error rates and gate error rates as well. Google's Sycamore Processor as well as IonQ's trapped-ion devices allow users to query qubit-level noise parameters as well. With minor modification, replacing the fake thermal data created for this project with live calibration data pulled from one of these devices could allow the algorithm to be used as a pre-compilation step in live quantum workflows.

6.2 Adaptive Thermal Remapping

A potential direction for future work is adaptive thermal remapping. The current methodology presented in this work requires qubit assignments to be fixed throughout circuit execution. However, allowing qubit assignments to change between executions based on updated calibration data could prove useful. Noise characteristics on quantum hardware can change rapidly — qubits can heat up and cool down, sometimes drastically, during the course of a day. An extension to this project that tracked stability scores over time and updated the mapping of logical to physical qubits accordingly could adapt to changes in device conditions without user intervention.

6.3 Integration with IBM Calibration Data

To that end, one specific direction for future work would be to integrate the mapper presented in this work directly with IBM's live calibration data. The Qiskit software development kit already has a built-in method of looking up the latest properties for any of IBM's backends, including per-qubit noise parameters. Swapping out the static CSV file used for thermal scores in this work for an API that pulls live data could automatically adjust physical qubit assignments to match device conditions at the time of circuit execution.

6.4 Learned Thermal-Aware Routing

Future work could also replace the greedy cost function described in this work with a reinforcement learning agent tasked with assigning qubits. Such an agent could theoretically learn the relationship between thermal scores, routing decisions, and circuit fidelity without human supervision. While the simple heuristic used in this work is likely insufficient to capture the large variation in device conditions across different hardware platforms, a learned agent could easily overcome this limitation with access to more training data. Training such an agent on a wide variety of circuits across many different devices could allow it to route more effectively than SABRE or the methodology presented here.

6.5 Applying Thermal-Aware Mapping to Larger Circuits

As noted in section 6, the implementation of this project currently uses an exhaustive search over all permutations of qubit mappings. While this approach is feasible for up to 5 qubits, it does not easily scale to larger circuits. Future work will have to consider alternative search strategies to make thermal-aware mapping practical for larger circuits. Possible algorithms could include graph partitioning, simulated annealing, or other types of heuristic search methods.

7. Conclusion

This project implemented a thermal aware qubit mapping algorithm and benchmarked it against SABRE and a random baseline. The primary contribution of this work was a method of adapting the SABRE routing algorithm to take device temperatures into account during the initial mapping stage. By defining a novel cost function that balances both routing overhead and thermal scores, the algorithm was able to route circuits with knowledge of device temperatures without changing any of the underlying routing methodology.

Experimental results confirmed the effectiveness of this approach, finding that thermal aware mapping outperforms both SABRE and the random mapping baseline in terms of circuit fidelity. While the increase in fidelity is modest, it does suggest that current quantum transpiler make suboptimal qubit placement decisions that lead directly to lower circuit fidelity. Unlike prior work which suggested hardware-aware routing could mitigate this problem [4], this project proposes a practical solution that can be deployed today to improve the fidelity of quantum circuits executed on real hardware.

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